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Physics of Living Systems

Microbiomes Through the Looking Glass

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This **important** study shows how the relative importance of inter-species interactions in microbiomes can be inferred from empirical species abundance data. The methods based on statistical physics of disordered systems are **convincing** and rigorous, and allow for distinguishing healthy and non-healthy human gut microbiomes via differences in their inter-species interaction patterns. This work should be of broad interest to researchers in microbial ecology and theoretical biophysics.

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Abstract

Bacterial communities are pivotal to maintaining ecological function and preserving the rich tapestry of biological diversity. The rapid development of environmental sequencing technologies, such as metagenomics, has revolutionized our capacity to probe such diversity. However, despite these advances, a theoretical understanding connecting empirical data with ecosystem modelling, in particular in the framework of disordered systems akin to spin glasses, is still in its infancy. Here, we present a comprehensive framework using theories of disordered systems to decode microbiome data, which offers insight into the ecological forces that shape macroecological states. By employing the quenched disordered generalized Lotka-Volterra model, we analyze species abundance data in healthy and diseased human gut microbiomes. Results reveal the emergence of two distinct patterns of species-interaction networks, elucidating the pathways through which dysbiosis may drive microbiome instability. Interaction patterns thus provide a window into the systemic shifts accompanying the transition from health to disease, offering a new perspective on the dynamics of the microbial community. Our findings suggest the potential of disordered systems theory to characterize microbiomes by capturing the essence of ecological interactions and their consequences on stability and functioning, leveraging our understanding of the linkages of dysbiosis and microbial dynamics.

1 Introduction

Bacterial communities are a fundamental reservoir of ecological functions and biological diversity. They are relevant for any environmental and host-associated ecosystem, ranging from soil to the human gut. In particular, the human gut microbiome, by interacting with host metabolism and immune system Levy et al. [2016], Barroso-Batista et al. [2015], Barreto and Gordo [2023], is a key regulator of human health. Dysbioses are defined as alterations of the healthy microbiome composition associated with the gastrointestinal tract Lloyd-Price et al. [2019], Das and Nair [2019], Nishida et al. [2018]. Some recent studies have shown that gut dysbioses are associated with changes in microbial species interaction patterns Bashan et al. [2016], Seppi et al. [2023], while emergent patterns focusing on single species properties Azaele et al. [2016], Grilli [2020], such as species abundance distribution (SAD), fail to characterize altered states of microbiome Seppi et al. [2023], Pasqualini et al. [2023]. Microbiomes host various interactions, measured for in-vitro communities Kehe et al. [2021], Venturelli et al. [2018], and are expected to govern natural ones.

Recently, stochastic non-interacting neutral Zeng et al. [2015], Venkataraman et al. [2015], Sala et al. [2016], Sieber et al. [2019] and logistic models Descheemaeker and De Buyl [2020], Grilli [2020], Zaoli and Grilli [2022], have been applied to give a satisfactory description of empirical patterns in different types of communities, ranging from human to soil microbiomes. However, they completely neglect species interactions in the population dynamics, and therefore they do not seem to have the right modelling tools to obtain novel insights from dysbioses models.

For these reasons, several approaches have been proposed in recent years to infer microbial interactions Faust and Raes [2012], Xiao et al. [2017], Camacho-Mateu et al. [2024]. However, such inference remains very challenging and problematic in many respects Faisal et al. [2010], Angulo et al. [2017], Tu et al. [2019], Armitage and Jones [2019], Holt [2020]: a) the very high dimension of typical microbiome datasets and the lack of longitudinal (long-term) experiments; b) the time-dependent nature of microbial species interactions Pacciani-Mori et al. [2021]; c) abiotic factors such as resources, temperature, and pH, can introduce environmental filtering effects Sireci et al. [2022] and induce effective interactions; d) experimental and technical challenges that introduce sampling effects, false positive species Tovo et al. [2020], spurious correlation, and bias in the data Weiss et al. [2016], Gloor et al. [2017], Dohlman and Shen [2019]. Therefore, even in a scenario where metagenomic samples are noise- and bias-free, reconstructing species-interaction networks of the underlying microbial dynamics remains a hard task.

An alternative, theoretically more elegant approach was pioneered by May May [1972], proposing to use random matrices to model species interaction networks, i.e., each entry of the adjacency matrix is extracted at random from a given distribution. Given the impossibility of empirically measuring the interaction strengths, the advantage of such an approach is the reduction from ~ S^2 (if S is the number of species) to just a few parameters (e.g., 2) used to parameterize the distribution Allesina and Tang [2012]. In recent years several investigations have applied this framework to the study of population dynamics through generalized Lotka-Volterra (gLV) equations, employing quenched disordered system techniques (also known as “glassy”), mutated from Statistical Physics Bunin [2017], Galla [2018], Biroli et al. [2018], Altieri et al. [2021], Lorenzana and Altieri [2022]. The main idea is thus to focus on the ensemble properties of the gLV, which will depend only on a few relevant control parameters (e.g., the mean μ and the variance σ^2 of the random interactions strengths) Barbier et al. [2018]. However, despite a range of very interesting results, this approach has remained confined mainly to purely theoretical domains (but see Hatton et al. [2024]) and has only recently been used to give an interpretation of controlled experiments with microbial communities Hu et al. [2022].

In this work, we therefore provide a *proof of concept* for the applicability of this framework to human microbiomes. In particular, we will infer the parameters of the quenched dgLV model from healthy and unhealthy cohorts of gut microbiomes. We will show that the different physiological states of the human gut microbiome correspond to different interaction patterns of the dgLV model. We will then introduce quantitative metrics to quantify the role of stability and the contribution of ecological forces to the dynamics.

2 Results

2.1 Disordered generalized Lotka-Volterra model

The dgLV model describes the evolution in time of the concentration abundances of a local pool of S interacting species, i.e.,

$$\frac{dN_i}{dt} = N_i \left[\rho_i (K_i - N_i) - \sum_{j,(j \neq i)} \alpha_{ij} N_j \right] + \eta_i(t) + \lambda \,, \tag{1}$$

where N_i is the populations of species i-th, K_i is its carrying capacity, and r_i the growth rate where $\rho_i = \frac{r_i}{K_i}$ sets the timescale of species dynamics, which we will assume will not depend on the species, i.e. $\rho_i = \rho$. The coefficients α_{ij} are random i.i.d random variables with $\mathbb{E}[\alpha_{ij}] = \mu/S$ and $\mathbb{E}[\alpha_{ij}]^2 - \mathbb{E}^2[\alpha_{ij}] = \sigma^2/S$. We also include an immigration species-independent rate λ, which will be treated as a reflecting wall mathematically regularizing the problem, and a demographic noise term Altieri et al. [2021] η_i, with $\langle \eta_i(t) \eta_j(t') \rangle = 2 N_i T \delta_{ij} \delta(t - t')$, and where δ denotes the Dirac delta and T sets the scale of noise intensity. For notational purposes, we also introduce its inverse $\beta = T^{-1}$. Finally, we require that the interactions are symmetric, i.e., $\alpha_{ij} = \alpha_{ji}$. This way, we can map this problem into an equilibrium thermodynamic one that is exactly solvable. As shown in the Supplementary Information (SI), section A, it is possible to justify this symmetric assumption and a linear dependence between the growth rate and the carrying capacity, $r_i = \rho K_i$, by considering the quasi-stationary approximation of the MacArthur consumer-resource model. In particular, we set $\rho = 1$ Biroli et al. [2018]. The presence of a noise term allows us to write the Fokker-Planck equation of the system and study its stationary solution (see Altieri et al. [2021], Altieri and Biroli [2022] for a detailed derivation in a similar Hamiltonian formalism). Here, we adopt the Itô prescription for the stochastic dynamics in such a way as to prevent species resurgence by noise Biroli et al. [2018].

The replica formalism, a well-known technique in disordered systems, comes into play and allows us to derive a non-interacting Hamiltonian corresponding to the dgLV symmetric interactions α_{ij} (see Methods and the SI, Section B for the complete derivation). In simple words, instead of trying to solve the problem for one random setup, the formalism considers many replicas of the system and averages their behaviors. This approach helps to smooth out the randomness and reveals the typical behavior of the system. We also assume a single equilibrium scenario, known as *replica-symmetric* (**RS**) regime. Although the SADs of empirical microbial communities display fat tails, a feature more compatible with the multi-attractors phase of the asymmetric 1RSB Mallmin et al. [2024] case or with the GLV with annealed disorder Suweis et al. [2023], we consider this simplification as it is the regime in which we can obtain explicit analytical relations between the model parameters and the data and where model inversion is feasible.

On the other hand, by using a cavity argument Mezard and Montanari [2009], Altieri and Baity-Jesi [2024], one can analytically derive the SAD of the model (see Methods and SI, Section D):

$$p(N|\zeta) \propto N^{\nu-1} \exp\left\{ -\beta \left(\frac{m}{2} N^2 - \zeta N \right) \right\}, \tag{2}$$

where the auxiliary variable $\zeta = K - \mu h + \sqrt{q_0}\sigma z$ takes the disorder interactions into account; z is a standardized Gaussian variable, $\nu = \beta\lambda > 0, h = \overline{\langle N \rangle}$, $q_d = \overline{\langle N^2 \rangle}$, $q_0 = \overline{\langle N \rangle^2}$, and $m = 1 - \beta\sigma^2(q_d - q_0)$ that should be greater than zero for the theory to be consistent Biroli et al. [2018].

The RS ansatz can also be characterized by its stability to external perturbations, such as the external temperature, the immigration rate, or the interaction heterogeneity. To investigate the stability of the RS phase, we consider the Hessian matrix of the theory by performing a harmonic expansion of the replicated free energy de Almeida and Thouless [1978]. When the leading eigenvalue of the Hessian matrix – suitably diagonalized on the so-called *replicon sector* – goes below zero, unstable equilibria appear: either the system moves towards a Replica Symmetric Breaking (RSB) phase with multiple locally stable minima, or it develops a marginal full-RSB phase, leading to a hierarchical structure of states and a very complex landscape (also known as glass phase). Therefore, in the latter cases, the RS ansatz no longer represents a thermodynamically stable phase. Based on standard calculations in disordered systems, this leading eigenvalue of the Hessian matrix, called *replicon R*, can be analytically evaluated as

$$\mathcal{R} = (\beta\sigma)^2 \left(1 - \sigma^2 \overline{\langle \left(\frac{\partial N}{\partial \xi} \right)^2 \rangle} \right) = (\beta\sigma)^2 \left(1 - (\beta\sigma)^2 \overline{(\langle N^2 \rangle - \langle N \rangle^2)^2} \right), \tag{3}$$

where $\frac{\partial N}{\partial \xi}$ is the species response to an external perturbation, keeping track of non-extinct species only. For more details, see SI Section C.

2.2 Data Through the Glass

We now consider cross-sectionally sampled gut microbiomes of two different cohorts: one of healthy and another one of diseased individuals. Here we focus on chronic inflammation syndromes, but, in principle, what we present holds for groups of phenotypically distinct populations. We can idealize the samples of each group as generated from the stationary distribution of the dgLV Eq.(1), with *different* realizations of disorder $\boldsymbol{\alpha}$, but with shared μ and ρ (see Methods). The approach we propose therefore considers all samples in the same group as coming from the same statistical ensemble but, on the other hand, it distinguishes ecological regimes characterized by distinct phenotypes through different statistics (e.g., dissimilar μ, and σ) of the random species interactions $\boldsymbol{\alpha}$. We aim to test the hypothesis that each cohort will be described by a different set of ecological parameters ($\boldsymbol{\theta} = (\mu, \sigma, T, \lambda)$).

To calculate the order parameters (h, q_d, q_0, K) from the data, we thus need to specify how we perform the ensemble and the disorder averages from the data (see **Figure 1**). Given that we typically lack time series data, and leveraging on an effective mean-field description, we consider estimating the ensemble average as the average among species (that is, different species are realizations of the same underlying stochastic process Azaele et al. [2016], known as neutral hypothesis), i.e., $\langle \cdot \rangle \sim \frac{1}{S} \sum_{species} \cdot$. On the other hand, we assume that the average over the disorder can be computed as a sample average. In other words, within a given phenotype (healthy/unhealthy) each measured microbiome configuration is a sample from the stationary distribution of the dgLV model, with a given realization of the disorder, i.e., $\overline{\cdot} \sim \frac{1}{R} \sum_{samples} \cdot$, where R is the number of samples and typically $R \gg 1$. Moreover, due to our limited knowledge of the

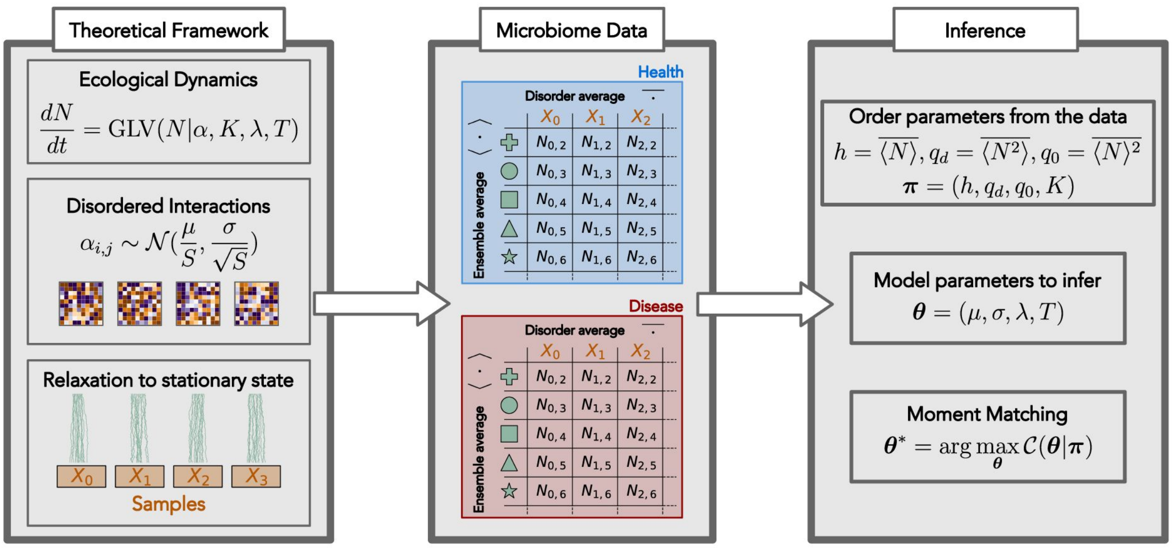

Figure 1

Inference of the dgLV generative model is performed by a moment matching optimization procedure. We aim to infer the free parameters $\boldsymbol{\theta} = (\mu, \sigma, T, \lambda)$ so that to match the order parameters and the average carrying capacity (h, q_0, q_d, K). This procedure allows us to extract ecological dynamics information for cross-sectional data of healthy (blue) and diseased (red) microbiomes, which are conceptualized as independent disorder realizations.

fine details governing the interactions in each microbiome, it is reasonable to assume that all communities with a given macrostate - a point in the (h, q_0, q_d, K) space - experience the same demographic noise T, immigration rate λ, and disorder parameters μ and σ. Eventually, we can compute the order parameters from the data as:

$$h = \overline{\langle N \rangle} = \frac{1}{R}\sum_{a=1}^{R}\left(\frac{1}{S}\sum_{j=1}^{S} N_{j,a}\right) ,$$

$$q_d = \overline{\langle N^2 \rangle} = \frac{1}{R}\sum_{a=1}^{R}\left(\frac{1}{S}\sum_{j=1}^{S} N_{j,a}^2\right) ,$$

$$q_0 = \overline{\langle N \rangle^2} = \frac{1}{R}\sum_{a=1}^{R}\left(\frac{1}{S}\sum_{j=1}^{S} N_{j,a}\right)^2$$

where $N_{a,j}$ gives the population density of species j in sample a.

To evaluate the carrying capacities K_i from the data, as done in most of the works using dGLV Bunin [2017], Biroli et al. [2018], Altieri et al. [2021], Suweis et al. [2023], Mallmin et al. [2024], we assume that in each cohort, all species are characterized by the same carrying capacity $K_i = K$. Here, we assign the value K as the average maximum relative abundance of species among the available samples for each cohort, thus $K = \frac{1}{S}\sum_{j=1}^{S} \max_a N_{j,a}$.

Note also that, from metagenomics, we only have access to compositional data for species abundances Pasqualini et al. [2023]. This is crucial to properly treat different samples and end up with a consistent analysis. Moreover, this also allows us to give an ecological interpretation of some of the order parameters. In particular, thanks to the compositionality of the data, we have that $h = \overline{S^{-1}}$ and $q_0 = \overline{S^{-2}}$ (where $\overline{S}$ is known as the average local diversity), while q_d provides information on pairwise products between species abundances within each replica. **Figure 2** shows the values of such order parameters between healthy and unhealthy cohorts with respect to two randomized cohorts. In the data, we observe a systematic difference between healthy and unhealthy cohorts, pointing to a higher average local diversity in healthy samples (panels A, B), as also observed in Pasqualini et al. [2023]. Panel C highlights the higher q_d in unhealthy patients, a signature of the weakening of the species interactions in those samples. We will investigate further this aspect in the following section, by inferring the species interactions in both cohorts.

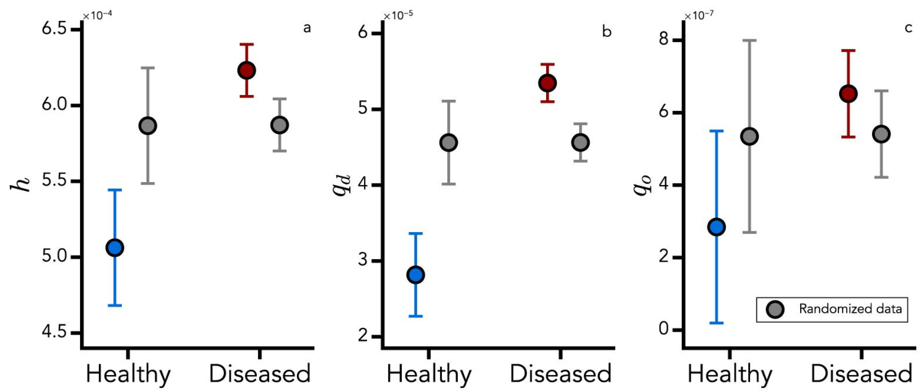

Figure 2

Order parameters inferred from the data using Eqs. (2.2). Panel A) h B) q_0 and C) q_d in healthy (blue) and unhealthy (red) cohorts. The gray points show the values for the corresponding null models where sample labels have been randomized. Standard deviations have been computed over 5000 realizations.

From the modelling point of view, it can be rigorously shown that constraining the abundances to be normalized to one corresponds to introducing an additional Lagrange multiplier in the expression of the free energy. However, this only acts on the linear term contributing to shifting the mean of the random interactions, μ, but does not have any influence on the heterogeneity σ, therefore on the phase diagram. For more details, we refer the interested reader to the SI (Section B1) along with Altieri et al. [2021]. In Altieri et al. [2021] the authors also showed how a global normalizing constraint on species abundances reflects in a one-to-one mapping with the random replicant model Biscari and Parisi [1995]. (The replicator equations originally introduced by Diederich and Opper [1989] and recast within the replica formalism Biscari and Parisi [1995], Altieri et al. [2021] describe the dynamics of an ensemble of replicants that evolve via random couplings.)

2.3 Species interactions patterns characterize the state of microbiomes

We thus collect all the parameters estimated from the data in a vector $\boldsymbol{\pi} = (h, q_d, q_0, K)$. As we will better detail in the Methods, we develop a moment matching inference algorithm to infer the model parameters $\boldsymbol{\theta}$. We therefore generated order parameters from our model that are as close as possible to the data, as depicted in **Figure 1**. The idea of the method is to introduce a cost function $C(\boldsymbol{\theta}|\boldsymbol{\pi})$, representing a total relative error for some self-consistent equations. If the parameters $\boldsymbol{\theta}$ are such that the right part of the self-consistent equation equals the left part, the problem can be considered solved. Since the landscape of this cost function presents several minima, we performed multiple optimization procedures to collect an ensemble of possible solutions, from which we retained the top 30.

First, we find that different solutions $\boldsymbol{\theta}^*$ of the optimization problem provides ecological insights into the underlying microbiome populations. As **Figure 3A** shows, the inferred demographic noise strength (T) and the heterogeneity of the interactions (σ) for healthy and diseased microbiomes are clustered in the T - σ plane. In particular, the SAD for the healthy cohort is robust among the different solutions of the inference procedure, as shown by the superposition of the different curves in the inset of **Figure 3A**. On the other hand, SADs inferred from unhealthy patients have high sensitivity to different solutions. In particular, some of them display a mode for high-abundance species (light red lines in **Figure 3A**), a signature of dominant strain in the gut. Consistently, the distribution of the interactions $P(\alpha_{i,j})$ generated through the inferred parameters μ and σ is different between healthy and diseased cohorts, giving a distinct pattern of interactions (see **Figure 3B**), a result that is compatible with that found by Bashan and collaborators Bashan et al. [2016]. In particular, we find that dysbiosis reduces the heterogeneity of interaction strengths, a result also observed when taking correlations as a proxy for interactions Seppi et al. [2023].

We then assess how close the inferred σ and $\beta = 1/T$ (the inverse temperature in a statistical physics approach) are to the critical replica symmetry-breaking (RSB) line of the dgLV ($R = 0$), evaluated being all the other parameters constant (see Methods). We find again that the replicon values R corresponding to each solution of our optimization protocol are significantly different for the two investigated microbiome phenotypes (see **Figure 4A**). In particular, diseased microbiomes are closer to marginal stability within the replica-symmetric ansatz Altieri et al. [2021], Mézard et al. [1987], de Almeida and Thouless [1978]. Furthermore, by investigating the shape of the SAD given by Eq. (2), we can estimate the ratio between niche (represented by species interaction) and neutral (represented by birth/death and immigration) ecological forces, which can be captured by the quantity ψ Wu et al. [2021]. It detects the emergence of internal modes in the species abundance distribution as a hallmark of niche processes (see SI, Section E).

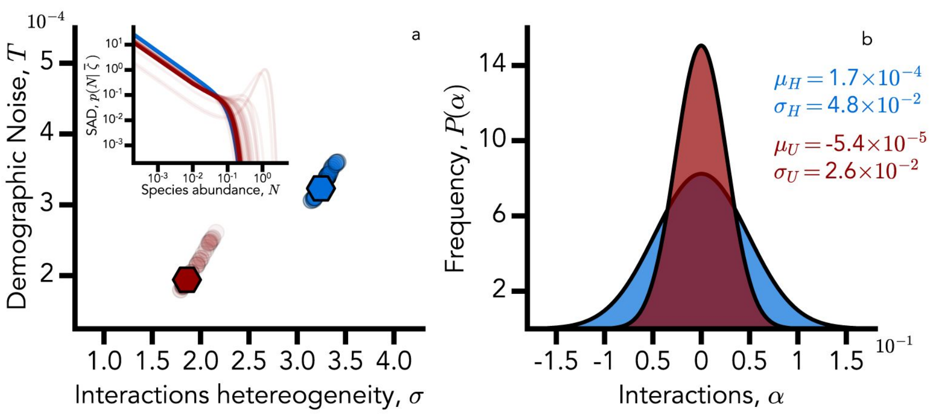

Figure 3

Panel a: inferred T (demographic noise strength) and σ (interactions heterogeneity) for healthy (blue) and diseased (red) microbiomes are clustered. Darker dots correspond to better solutions (i.e., solutions with a lower value of the cost function C), while the two points with hexagonal markers correspond to the best two (healthy and diseased, respectively) solutions. In the first panel inset, we also show (in log-log scale) the SADs corresponding to each solution. To have a more concise representation, we present each SAD fixing the disorder to its average $\bar{\zeta} = K - \mu h$. Panel b: the probability density function of the inferred interactions $\alpha_{i,j}$ for healthy (blue) and diseased (red) microbiomes. Dysbiosis reduces the heterogeneity of the interaction strengths.

Inspired by field-theory arguments (see Methods section *Free-energy landscape exploration: replica formulation*), we can call *mass* of the theory the m parameter as defined above in Eq. (2). When $m \to 0$ the theoretical order parameters diverge and the system enters a pathological regime of unbounded growth. As such, the *mass* term, can be considered as a complementary stability measure, capable of capturing the transition to the unbounded growth regime.

In the model, two kinds of effects compete to shape the community structure. On the one hand, we have niche effects, encoded in disordered interactions and thus tracked by the parameters μ, σ, and K. Their overall effect is selective and tends to concentrate the SAD around the typical abundance value. On the other hand, we have neutral effects encoded into the stochastic dynamics and immigration, governing the low abundance regime of the SAD. When the demographic noise amplitude is stronger than immigration ($\nu < 1$, as in our case) the SAD exhibits a low-abundance integrable divergence. In the opposite scenario, $\nu > 1$, there is no divergence and the SAD is modal. Since interactions are random, the probability of observing an internal mode can be estimated as the fraction of SADs realization having non-trivial solutions to the stationary point equation. Such quantity, dubbed as the niche-neutral ratio, can be analytically evaluated and reads:

$$\psi = \frac{1}{2}\mathrm{Erfc}\left(\frac{\zeta^* + \overline{\zeta}}{\sqrt{2}\sigma_\zeta}\right) + \frac{1}{2}\mathrm{Erfc}\left(\frac{\zeta^* - \overline{\zeta}}{\sqrt{2}\sigma_\zeta}\right) , \tag{5}$$

where $\zeta^* = \sqrt{\frac{4(1-\nu)m}{\beta}}$ and $\overline{\zeta} = K - \mu h$. When $\psi \approx 1$, niche and neutral forces give comparable contributions to the dynamics, as both low abundance divergence and a finite abundance mode coexist in the SAD. Finally, if the typical abundance diverges, we enter into the unbounded growth phase, which means that the mass m and the niche-neutral ratio ψ are not independent, as suggested by the analytical expression on ψ. For an exhaustive derivation of this result, see SI (Section E). With the model parameters obtained, we were able to evaluate m and ψ for healthy and diseased microbiomes. Also, in this case, healthy and diseased microbiomes are clearly clustered, as shown in **Figure (4)**. Unhealthy microbiomes turn out to be closer to the unbounded growth phase, and the nice-neutral ratio is larger by five orders of magnitude compared to the healthy case $\psi_U \approx 10^5\psi_H$. This leads us to argue that selective pressure is way larger in diseased states, while in the healthy one, birth and death effects are the key drivers of the dynamics. These results are also confirmed by the SADs shape in the inset of **Figure (3**, panel a).

Discussion and Conclusions

In our exploration of the gut microbiome through the lens of disordered systems and random matrix theory, we have embarked on an ambitious journey to connect intricate theoretical concepts to practical analyses of environmental microbiome data. In particular, we have characterized healthy and unhealthy gut microbiomes using the disordered generalized Lotka-Volterra model (dgLV) for population dynamics. This attempt to apply advanced theoretical methodologies from disordered systems to real-world biological data represents a significant stride in understanding the complex interplay within microbial communities.

One delicate aspect of our study lies in addressing the inherent limitations and the potential scope for improvement in future research. A notable challenge is the discrepancy between the empirical Species Abundance Distribution (SAD), which is observed in the data, and the theoretical SAD predicted by the quenched dgLV model Suweis et al. [2023]. While empirical data showcase a diverse range of species abundance, following a power-law distribution, the model predictions tend to exhibit exponential decay in SAD tails. This mismatch underscores the need for further refinement of the model to accurately capture the nuanced patterns observed in real-world data. For example, recently it has been proposed that introducing an annealed disorder can generate

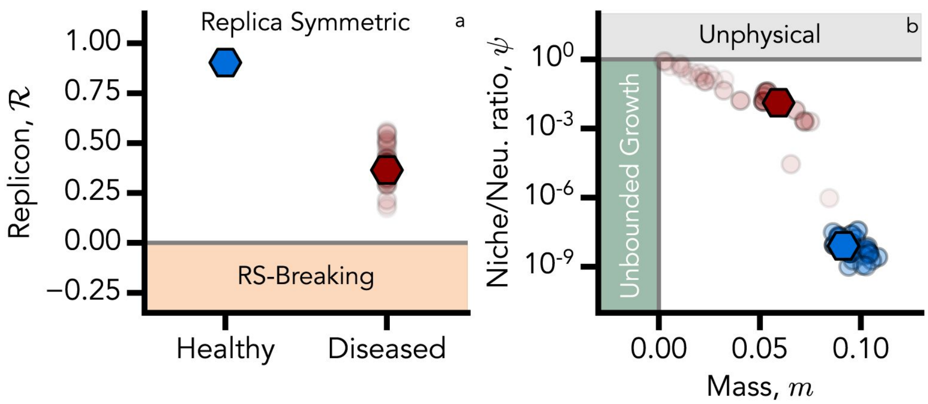

Figure 4

The replicon eigenvalue corresponding to each solution of our optimization procedure (shaded dots). The solid hexagon represents the replicon corresponding to the best solutions that minimize the error in predicting the order parameters of the theory (minimum *C*). The two investigated microbiome phenotypes (healthy in blue, diseased in red) are significantly different. In particular, diseased microbiomes are closer to the marginal stability of replica-symmetric ansatz (grey horizontal line). Panel b: Solutions of the moment-matching objective function are shown as a function of ψ and m, which in turn depend on the SAD parameters (see main text). Healthy (blue) and diseased (red) microbiomes appear to be clustered. Therefore, distinct ecological organization scenarios (strong neutrality/emergent neutrality) emerge. Darker dots correspond to solutions with lower values of the cost function, while hexagonal markers correspond to the two best solutions.

SADs that look closer to the empirical ones Suweis et al. [2023]. Another possibility is to set the dgLV model parameters in such a way as to reproduce the multi-attractor phase. In fact, in this region, the SADs display a more heterogeneous shape Arnoulx de Pirey and Bunin [2024]. We already intend to explore this follow-up direction by combining one-step replica-symmetry-breaking (1RSB) computations in the multiple-attractor phase with Dynamical Mean-Field Theory analysis for the asymmetric interaction case.

Another related limitation is the challenge of generating species abundance samples from the dgLV model that mirrors the statistical properties of the observed empirical data. In our current framework, each microbiome sample could be extracted from $p(N|\zeta)$, where ζ is a realization of the disorder. However, $p(N|\zeta)$ near $N \to 0$ presents a power-law exponent ν - 1, with $\nu_{H,D} \approx 10^{-3}$. This generates numerical instabilities and dominates the sampling process, posing difficulties in generating representative synthetic samples. Moreover, while microbiome data are inherently compositional Pasqualini et al. [2023], the dgLV model species populations N_i are positive real numbers. However, as already noted, it can be shown that introducing a normalization $\tilde{N}_i \to N_i / \sum_j N_j$ in such equations does not change the structure of the proposed solutions Altieri et al. [2021] and therefore should not affect the conclusions of our work. The study also sheds light on the role of demographic noise within the context of the dgLV model. In fact, in the limit $T \to 0$, the SAD transitions to a truncated Gaussian, with a prominent peak at zero reflecting the fraction of extinct species. In contrast to scenarios at zero noise, where species extinctions are observed, the inclusion of demographic noise in the dgLV model suggests a scenario where no species goes extinct, supporting the *everything is everywhere* hypothesis Grilli [2020], Pigani et al. [2024]. In other words, within this framework, zeros in the data are due to sampling effects and not to local species extinctions Grilli [2020], Pasqualini et al. [2023].

In conclusion, our work opens up new avenues for future research, particularly in refining the theoretical models to better align with empirical observations and in exploring the nuances of species abundance distributions within the microbiome context. Moreover, the integration of other forms of environmental variability and species-specific traits could provide a more holistic view of ecosystem dynamics as also proposed by the *One Health-One Microbiome* framework Tomasulo et al. [2024]. The insights gained from this study not only enhance our understanding of the complex dynamics in microbial ecosystems but also pave the way for more sophisticated analytical tools in ecological modelling, contributing significantly to our ability to interpret and manage these vital biological systems.

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Reviewer #1 (Public review):

Summary:

In this manuscript, the authors develop a novel method to infer ecologically-informative parameters across healthy and diseased states of the gut microbiota, although the method is generalizable to other datasets for species abundances. The authors leverage techniques from theoretical physics of disordered systems to infer different parameters - mean and standard deviation for the strength of bacterial interspecies interactions, a bacterial immigration rate, and the strength of demographic noise - that describe the statistics of microbiota samples from two groups-one for healthy subjects and another one for subjects with chronic inflammation syndromes. To do this, the authors simulate communities with a modified version of the Generalized Lotka-Volterra model and randomly-generated interactions, and then use a moment-matching algorithm to find sets of parameters that better reproduce the data for species abundances. They find that these parameters are different for the healthy and diseased microbiota groups. The results suggest, for example, that bacterial interaction strengths, relative to noise and immigration, are more dominant for microbiota dynamics in diseased states than in healthy states.

We think that this manuscript brings an important contribution that will be of interest in the areas of statistical physics, (microbiota) ecology, and (biological) data science. The evidence of their results is solid and the work improves the state-of-the-art in terms of methods. There are a few weaknesses that, in our opinion, the authors could address to further improve the work.

Strengths:

(1) Using a fairly generic ecological model, the method can identify the change in the relative importance of different ecological forces (distribution of interspecies interactions, demographic noise, and immigration) in different sample groups. The authors focus on the case of the human gut microbiota, showing that the data are consistent with a higher influence of species interactions (relative to demographic noise and immigration) in a disease microbiota state than in healthy ones.

(2) The method is novel, original, and it improves the state-of-the-art methodology for the inference of ecologically relevant parameters. The analysis provides solid evidence for the conclusions.

Weaknesses:

In the way it is written, this work might be mostly read by physicists. We believe that, with some rewriting, the authors could better highlight the ecological implications of the results and make the method more accessible to a broader audience.

https://doi.org/10.7554/eLife.105948.1.sa3

Reviewer #2 (Public review):

Summary:

This valuable work aims to infer, from microbiome data, microbial species interaction patterns associated with healthy and unhealthy human gut microbiomes. Using solid techniques from statistical physics, the authors propose that healthy and unhealthy microbiome interaction patterns substantially differ. Unhealthy microbiomes are closer to instability and single-strain dominance; whereas healthy microbiomes showcase near-neutral dynamics, mostly driven by demographic noise and immigration.

Strengths:

A well-written article, relatively easy to follow and transparent despite the high degree of technicality of the underlying theory. The authors provide a powerful inferring procedure, which bypasses the issue of having only compositional data.

Weaknesses:

(1) This sentence in the introduction seems key to me: "Focusing on single species properties as species abundance distribution (SAD), fail to characterise altered states of microbiome." Yet it is not explained what is meant by 'fail', and thus what the proposed approach 'solves'.

(2) Lack of validation, following arbitrary modelling choices made (symmetry of interactions, weak-interaction limit, uniform carrying capacity).
Inconsistent interpretation of instability. Here, instability is associated with the transition to

the marginal phase, which becomes chaotic when interaction symmetry is broken. But as the authors acknowledge, the weak interaction limit does not reproduce fat-tailed abundance distributions found in data. On the other hand, strong interaction regimes, where chaos prevails, tend to do so (Mallmin et al, PNAS 2024). Thus, the nature of the instability towards which unhealthy microbiomes approach is unclear.

(3) Three technical points about the methodology and interpretation.
a) How can order parameters h and q0 can be inferred, if in the compositional data they are fixed by definition?
b) How is it possible that weaker interaction variance is associated with approach to instability, when the opposite is usually true?
c) Having an idea of what the empirical data compares to the theoretical fits would be valuable.

Implications:

As the authors say, this is a proof of concept. They point at limits and ways to go forward, in particular pointing at ways in which species abundance distributions could be better reproduced by the predicted dynamical models. One implication that is missing, in my opinion, is the interpretability of the results, and what this work achieves that was missing from other approaches (see weaknesses section above): what do we learn from the fact that changes in microbial interactions characterise healthy from unhealthy microbiota? For instance, what does this mean for medical research?

https://doi.org/10.7554/eLife.105948.1.sa2

Reviewer #3 (Public review):

Summary:

I found the manuscript to be well-written. I have a few questions regarding the model, though the bulk of my comments are requests to provide definitions and additional clarity. There are concepts and approaches used in this manuscript that are clear boons for understanding the ecology of microbiomes but are rarely considered by researchers approaching the manuscript from a traditional biology background. The authors have clearly considered this in their writing of S1 and S2, so addressing these comments should be straightforward. The methods section is particularly informative and well-written, with sufficient explanations of each step of the derivation that should be informative to researchers in the microbial life sciences who are not well-versed with physics-inspired approaches to ecology dynamics.

Strengths:

The modeling efforts of this study primarily rely on a disordered form of the generalized Lotka-Volterra (gLV) model. This model can be appropriate for investigating certain systems, and the authors are clear about when and how more mechanistic models (i.e., consumer-resource) can lead to gLV. Phenomenological models such as this have been found to be highly useful for investigating the ecology of microbiomes, so this modeling choice seems justified, and the limitations are laid out.

Weaknesses:

The authors use metagenomic data of diseased and healthy patients that were first processed in Pasqualini et al. (2024). The use of metagenomic data leads me to a question regarding the role of sampling effort (i.e., read counts) in shaping model parameters such as h. This parameter is equal to the average of 1/# species across samples because the data are compositional in nature. My understanding is that h was calculated using total abundances (i.e., read counts). The number of observed species is strongly influenced by sampling effort, so it would be useful if the number of reads were plotted against the number of species for healthy and diseased subjects.

However, the role of sampling effort can depend on the type of data, and my instinct about the role that sampling effort plays in species detection is primarily based on 16S data. The dependency between these two variables may be less severe for the authors' metagenomic pipeline. This potential discrepancy raises a broader issue regarding the investigation of microbial macroecological patterns and the inference of ecological parameters. Often microbial macroecology researchers rely on 16S rRNA amplicon data because that type of data is abundant and comparatively low-cost. Some in microbiology and bioinformatics are increasingly pushing researchers to choose metagenomics over 16S. Sometimes this choice is valid (discovery of new MAGs, investigate allele frequency changes within species, etc.), sometimes it is driven by the false equivalence "more data = better". The outcome, though, is that we have a body of more-or-less established microbial macroecological patterns which rest on 16S data and are now slowly incorporating results from metagenomics. To my knowledge, there has not been a systematic evaluation of the macroecological patterns that do and do not vary by one's choice in 16S vs. metagenomics. Several of the authors in this manuscript have previously compared the MAD shape for 16S and metagenomic datasets in Pasqualini et al., but moving forward, a more comprehensive study seems necessary (2024).

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Author response:

> ***Reviewer #1:***
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> *Strengths:*
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> *(1) Using a fairly generic ecological model, the method can identify the change in the relative importance of different ecological forces (distribution of interspecies interactions, demographic noise, and immigration) in different sample groups. The authors focus on the case of the human gut microbiota, showing that the data are consistent with a higher influence of species interactions (relative to demographic noise and immigration) in a disease microbiota state than in healthy ones. (2) The method is novel, original, and it improves the state-of-the-art methodology for the inference of ecologically relevant parameters. The analysis provides solid evidence for the conclusions.*
>
> *Weaknesses:*
>
> *In the way it is written, this work might be mostly read by physicists. We believe that, with some rewriting, the authors could better highlight the ecological implications of the results and make the method more accessible to a broader audience.*

We thank the reviewer for their positive and constructive feedback. We particularly appreciate the recognition of the novelty and robustness of our method, as well as the insight that it sheds light on the shifting ecological forces between healthy and diseased microbiomes. In response to the concern about the manuscript's accessibility, we aim to revise key sections – including the Introduction, Results, and Discussion – to more clearly articulate the ecological relevance of our theoretical findings. We would like to emphasize that our approach offers a novel perspective for analyzing individual species' abundances, as well as for understanding interaction patterns and stability at the community level. By placing our results within a broader context accessible to readers from diverse backgrounds, we aim for the revised version to appeal to a wider audience, including ecologists and microbiome scientists, while preserving the rigor of our underlying statistical physics framework.

> ***Reviewer #2:***
>
> *Strengths:*
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> *A well-written article, relatively easy to follow and transparent despite the high degree of technicality of the underlying theory. The authors provide a powerful inferring procedure, which bypasses the issue of having only compositional data.*
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> *Weaknesses:*
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> *(1) This sentence in the introduction seems key to me: "Focusing on single species properties as species abundance distribution (SAD), it fails to characterise altered states of microbiome." Yet it is not explained what is meant by 'fail', and thus what the proposed approach 'solves'. (2) Lack of validation, following arbitrary modelling choices made (symmetry of interactions, weak-interaction limit, uniform carrying capacity). Inconsistent interpretation of instability. Here, instability is associated with the transition to the marginal phase, which becomes chaotic when interaction symmetry is broken. But as the authors acknowledge, the weak interaction limit does not reproduce fat-tailed abundance distributions found in data. On the other hand, strong interaction regimes, where chaos prevails, tend to do so (Mallmin et al, PNAS 2024). Thus, the nature of the instability towards which unhealthy microbiomes approach is unclear. (3) Three technical points about the methodology and interpretation. a) How can order parameters h and $q0$ can be inferred, if in the compositional data they are fixed by definition? b) How is it possible that weaker interaction variance is associated with an approach to instability, when the opposite is usually true? c) Having an idea of what the empirical data compares to the theoretical fits would be valuable. Implications: As the authors say, this is a proof of concept. They point at limits and ways to go forward, in particular pointing at ways in which species abundance distributions could be better reproduced by the predicted dynamical models. One implication that is missing, in my opinion, is the interpretability of the results, and what this work achieves that was missing from other approaches (see weaknesses section above): what do we learn from the fact that changes in microbial interactions characterise healthy from unhealthy microbiota? For instance, what does this mean for medical research?*

We greatly appreciate the reviewer's thoughtful analysis highlighting both the strengths and areas of ambiguity in our work.

(1) To clarify the sentence on the limitations of species abundance distributions (SADs), we aim to explain in the revised version that while SADs summarize the relative abundance of individual species, they fail to capture the species-species correlations that we have shown (Seppi et al., Biomolecules 2023) to be more susceptible to the healthy state of the host. Our method thus focused on the interaction statistics among species, providing insights into underlying dynamics and stability of the microbiomes and their differences between healthy and unhealthy hosts.

(2) Regarding model assumptions, we acknowledge that the weak interaction regime and symmetry hypotheses simplify the analysis and may not capture all empirical richness, such

as fat-tailed distributions of species abundance. However, we interpret instability not as a path to chaos per se, but as a transition toward a multi-attractor phase, where each microbiome reaches a different fixed point. This is consistent with prior empirical findings invoking the "Anna Karenina principle", where healthy microbiomes resemble one another, but disease states tend to deviate from this picture (see Pasqualini et al., PLOS Comp. Bio. 2024). We consider our framework as a starting point and agree that further extensions incorporating strong interaction regimes (as suggested by Mallmin et al., PNAS 2024) or relaxing other model assumptions could reveal even richer dynamical patterns. The computational pipeline we present can be, in fact, easily generalizable to include different population dynamics models.

On the technical questions: (a) While compositional data constrain relative abundances, we can still estimate diversity-dependent parameters (h and q0) using alpha-diversity statistics across samples, which show meaningful variation; (b) The counter-intuitive instability that the reviewer pointed out arises from the interplay between demographic stochasticity and quenched disorder. It is the combined contribution of these two factors in phase space – not either one alone – that drives the transition. For clarity, see Figure 1 in Altieri et al., Phys. Rev. Lett. 2021; (c) We plan to include plots that compare empirical data to theoretical model fits. This will help visualize how well the model captures observed microbial community properties demographic noise (T), healthy communities are more stable (i.e., distantσ from the and how even with larger species interaction heterogeneity (σ) and larger critical line), as measured, by the replicon eigenvalue. Finally, regarding interpretability and implications: by showing that ecological interaction networks – not just species identities – differ between healthy and unhealthy states, our work suggests a conceptual shift. This could inform medical strategies aimed at restoring community-level stability rather than targeting individual microbes. In the revised Discussion section, we will elaborate on this point to better highlight its practical implications and outline potential directions for future research.

> ***Reviewer #3:***
>
> *Strengths:*
>
> *The modeling efforts of this study primarily rely on a disordered form of the generalized Lotka-Volterra (gLV) model. This model can be appropriate for investigating certain systems, and the authors are clear about when and how more mechanistic models (i.e., consumer-resource) can lead to gLV. Phenomenological models such as this have been found to be highly useful for investigating the ecology of microbiomes, so this modeling choice seems justified, and the limitations are laid out.*
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> *and bioinformatics are increasingly pushing researchers to choose metagenomics over 16S. Sometimes this choice is valid (discovery of new MAGs, investigate allele frequency changes within species, etc.), sometimes it is driven by the false equivalence "more data = better". The outcome, though, is that we have a body of more-or-less established microbial macroecological patterns which rest on 16S data and are now slowly incorporating results from metagenomics. To my knowledge, there has not been a systematic evaluation of the macroecological patterns that do and do not vary by one's choice in 16S vs. metagenomics. Several of the authors in this manuscript have previously compared the MAD shape for 16S and metagenomic datasets in Pasqualini et al., but moving forward, a more comprehensive study seems necessary.*

We thank the reviewer for this insightful and nuanced comment, which particularly highlights the broader methodological context of our data sources. Indeed, metagenomic sequencing introduces different biases with respect to 16S data. First, we would like to emphasize that we estimated the order parameters from the data by using relative abundances. Second, while the concern regarding the influence of sequencing depth and species diversity on the estimation of the order parameters is valid, we refer to a previous publication by some of the authors (Pasqualini et al., 2024; see Figure 4, panels g and h). There, we pointed out that the observed outcome is weakly influenced by sequencing depth in our dataset, while the main impact on the order parameters estimate comes from the species diversity of the two groups. In the same publication, we showed that other well-known patterns (species abundance distribution, mean abundance distribution) are also observed. Also, to mitigate the effect of the number of samples and sequencing depth, we estimated the order parameters by a bootstrap procedure (90% of samples for healthy and diseased groups, 5000 resamples), which resulted in the error bars in Figure 2.

We also fully agree with the broader call for a systematic comparison of macroecological patterns derived from 16S and metagenomic data. While some of us have already begun exploring this direction (e.g., Pasqualini et al., 2024), the reviewer's comment highlights its significance and motivates us to pursue a more comprehensive, integrative analysis across data types. While we found qualitative agreement of these patterns with previous publications (e.g., Grilli, Nature Comm. 2020), we will acknowledge this as an important future direction in the Discussion section.

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